

Multiple Choice (10 marks)

1. Higher dietary sodium (salt) intake is generally associated with:
 - (a) Decreased calcium excretion in the urine
 - (b) Increased risk of fracture
 - (c) Decreased dietary calcium absorption
 - (d) Negative calcium balance and bone mineral loss
2. The daily amount of protein oxidized in the body of a person spending 24 hours in a respiratory chamber can be determined from his/her:
 - (a) O₂ consumption alone
 - (b) CO₂ production alone
 - (c) Breath collection
 - (d) Total urinary nitrogen excretion alone
3. According to the recommendations for the nutritional management of patients with diabetes, the consumption of saturated fat should be:
 - (a) <10% of total daily energy
 - (b) <12%
 - (c) <15%
 - (d) <16%
4. What is the phenotype of a congenital disorder impairing the secretion of leptin?
 - (a) Normal energy intake, normal body weight and hyperthyroidism
 - (b) Obesity, excess energy intake, normal growth and hypoinsulinaemia
 - (c) Obesity, abnormal growth, hypothyroidism, hyperinsulinaemia
 - (d) Underweight, abnormal growth, hypothyroidism, hyperinsulinaemia
5. What enzyme is responsible for copying methylation marks from the parental to the daughter strand of DNA during replication?
 - (a) DNMT 1
 - (b) DNMT 3 a
 - (c) DNMT 3 b
 - (d) DNMT 3 L

True / False (10 marks)

Answer True or False

6. True or False: Starch mainly contains amylose, which is a polymer of glucose with alpha 1,4-linkages.
7. True or False: C₁₈ and C₂₂ polyunsaturated fatty acids are the primary precursors for eicosanoid synthesis.
8. True or False: Supplementation with EPA and DHA during pregnancy is primarily beneficial for improving memory in infants.
9. True or False: Saturated and trans unsaturated fatty acids are known to improve plasma LDL cholesterol levels.
10. True or False: The muscle fibre is the basic functional unit of skeletal muscle tissue and is responsible for muscle contraction.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how increased insulin secretion in the fed state influences tissue protein synthesis and analyze its significance in overall health and nutrition.
12. Discuss how high doses of β -carotene supplements may influence the risk of developing lung cancer, considering the mechanisms and existing research findings.

End of Paper